PART A — MODULE 1 QUESTIONS

MATH7-M1-001
Prompt:
What is the solution to $(2/3)x - 5 = 7$?
A) 8
B) 12
C) 18
D) 24

MATH7-M1-002
Prompt:
A machine fills 18 bottles in 6 minutes at a constant rate. At this rate, how many bottles does the machine fill in 11 minutes?
A) 24
B) 30
C) 33
D) 36

MATH7-M1-003
Prompt:
The numbers of text messages sent by 5 students in one hour are 12, 18, 15, 21, and 14. What is the median number of text messages sent?
A) 14
B) 15
C) 16
D) 18

MATH7-M1-004
Prompt:
A line in the xy-plane has equation $y = 2x - 9$. What is the slope of the line?
A) -9
B) -2
C) 2
D) 9

MATH7-M1-005
Prompt:
A student needs at least 85 points to pass a course. The student has already earned 52 points and will earn p more points. Which inequality represents this situation?
A) $p + 52 \geq 85$
B) $p + 52 \leq 85$
C) $52p \geq 85$
D) $85 - p \geq 52$

MATH7-M1-006
Prompt:
What is the solution (x, y) to the system of equations?
$x + 2y = 16$
$x = 4$
A) (4, 4)

B) (4, 6)
C) (6, 4)
D) (8, 4)

MATH7-M1-007
Prompt:
Which expression is equivalent to $9z - 4(2z - 3)$?
A) $z - 12$
B) $z + 12$
C) $17z - 12$
D) $17z + 12$

MATH7-M1-008
Prompt:
Which expression is equivalent to $b^8 \div b^2$, where $b \neq 0$?
A) b^4
B) b^6
C) b^{10}
D) $2b^6$

MATH7-M1-009
Prompt:
A bowl contains 7 almonds, 9 cashews, and 4 walnuts. If one nut is selected at random, what is the probability that the nut selected is not a walnut?
A) 1/5
B) 2/5
C) 4/5
D) 9/20

MATH7-M1-010
Prompt:
A triangle has an area of 54 square inches and a base of 12 inches. What is the height, in inches, of the triangle?
A) 4.5
B) 6
C) 9
D) 18

MATH7-M1-011
Prompt:
In the xy-plane, point A has coordinates $(-4, 5)$, and point B has coordinates $(2, 5)$. What is the x-coordinate of the midpoint of segment AB?
A) -2
B) -1
C) 1
D) 3

MATH7-M1-012
Prompt:
What is the value of $\sqrt{100} - \sqrt{36}$?
A) 2

B) 4
C) 6
D) 8

MATH7-M1-013
Prompt:
If $f(x) = 4x - 1$ and $f(t) = 23$, what is the value of t?
A) 5
B) 6
C) 7
D) 8

MATH7-M1-014
Prompt:
Which of the following is a solution to $x^2 = 49$?
A) −8
B) −7
C) 6
D) 14

MATH7-M1-015
Prompt:
A backpack costs $50 before a 12% discount is applied. What is the price of the backpack after the discount?
A) $38
B) $42
C) $44
D) $48

MATH7-M1-016
Prompt:
The table shows the relationship between the number of identical packets and the total number of stickers.

Packets: 3, 5, 8, 10
Stickers: 24, 40, 64, 80

How many stickers are in 1 packet?
A) 6
B) 8
C) 10
D) 12

MATH7-M1-017
Prompt:
A circle has diameter 14 centimeters. What is the circumference, in centimeters, of the circle?
A) 7π
B) 14π
C) 28π
D) 49π

MATH7-M1-018
Prompt:
The surface area S of a cube with side length a is $S = 6a^2$. What is the surface area of a cube with side length 4 inches?
A) 24
B) 64
C) 96
D) 144

MATH7-M1-019
Prompt:
An equation is given by $T = 18 + 4d$, where T is the total number of practice problems completed after d days. What does 18 represent in this equation?
A) The number of practice problems completed each day
B) The number of days spent practicing
C) The number of practice problems already completed before the d days
D) The total number of practice problems completed after 18 days

MATH7-M1-020
Prompt:
The ratio of green beads to purple beads in a bracelet is 2:7. If there are 18 beads total, how many are purple?
A) 4
B) 7
C) 12
D) 14

MATH7-M1-021
Prompt:
A rectangular floor has length 13 feet and width 9 feet. What is the area, in square feet, of the floor?
A) 22
B) 44
C) 117
D) 169

MATH7-M1-022
Prompt:
The table shows the number of hours that several students spent studying for a test.

Hours studied: 1, 2, 3, 4
Number of students: 3, 6, 5, 2

How many students are represented in the table?
A) 10
B) 12
C) 14
D) 16

PART B — MODULE 2 QUESTIONS

MATH7-M2-001
Prompt:
What is the solution to $4 - (x - 3)/2 = 9$?
A) −13
B) −7
C) 7
D) 13

MATH7-M2-002
Prompt:
The system of equations has solution (x, y).
$4x - y = 19$
$x + 3y = 18$
What is the value of $y - x$?
A) −2
B) 1
C) 3
D) 5

MATH7-M2-003
Prompt:
The function f is defined by $f(x) = x^2 - 4x - 12$. What is the minimum value of $f(x)$?
A) −16
B) −12
C) −8
D) 4

MATH7-M2-004
Prompt:
Which expression is equivalent to $(x^2 - 9)/(x^2 + 3x)$, where $x \neq 0$ and $x \neq -3$?
A) $1/x$
B) $(x - 3)/x$
C) $(x + 3)/x$
D) $x - 3$

MATH7-M2-005
Prompt:
A population of insects decreases by 25% each week. If the starting population is 1,600 insects, which expression gives the population after 3 weeks?
A) $1{,}600(0.25)^3$
B) $1{,}600(0.75)^3$
C) $1{,}600 - 3(0.25)$
D) $1{,}600 - 3(0.75)$

MATH7-M2-006
Prompt:
A number is increased by 40%, and the result is 84. What was the original number?
A) 48
B) 60
C) 70
D) 118

MATH7-M2-007
Prompt:
The table shows the numbers of students in two grade levels who chose different elective courses.

Elective: Art, Coding, Theater
Grade 10: 18, 22, 10
Grade 11: 12, 28, 20

If one of these students is chosen at random, what is the probability that the student chose Coding?
A) 1/4
B) 5/11
C) 1/2
D) 6/11

MATH7-M2-008
Prompt:
A right rectangular prism has length 9 inches, width 4 inches, and height 7 inches. What is the total surface area, in square inches, of the prism?
A) 126
B) 172
C) 218
D) 252

MATH7-M2-009
Prompt:
In a right triangle, $\tan \theta = 7/24$. What is $\sin \theta$?
A) 7/25
B) 7/24
C) 24/25
D) 25/24

MATH7-M2-010
Prompt:
A line passes through the points $(-2, 10)$ and $(6, -6)$. What is the x-intercept of the line?
A) 2
B) 3
C) 4
D) 5

MATH7-M2-011
Prompt:
The functions f and g are defined by $f(x) = 3x^2$ and $g(x) = x + 4$. What is $f(g(-1))$?
A) 9
B) 18
C) 27
D) 36

MATH7-M2-012
Prompt:
What is the greater solution to $|x + 1| = 8$?
A) −9

B) -7
C) 7
D) 9

MATH7-M2-013
Prompt:
A circle in the xy-plane has equation $(x + 5)^2 + (y - 1)^2 = 36$. What is the radius of the circle?
A) 1
B) 5
C) 6
D) 36

MATH7-M2-014
Prompt:
For what value of c does the equation $5x - 2 = cx - 2$ have infinitely many solutions?
A) -5
B) -2
C) 2
D) 5

MATH7-M2-015
Prompt:
The table shows values of a function h.

x: $-2, -1, 0, 1$
h(x): $-6, 0, 2, 0$

Which equation could define h?
A) $h(x) = -2x^2 - 2x + 2$
B) $h(x) = -2x^2 + 2$
C) $h(x) = 2x^2 + 2x - 2$
D) $h(x) = 2x^2 - 2$

MATH7-M2-016
Prompt:
Which expression is equivalent to $(3x^2 - 12x)/(6x)$, where $x \neq 0$?
A) $(x - 4)/2$
B) $(x - 12)/6$
C) $x - 2$
D) $3x - 12$

MATH7-M2-017
Prompt:
A data set has 6 values with a mean of 11. A value of 23 is added to the data set. What is the mean of the new data set?
A) 12
B) 12.5
C) 13
D) 13.5

MATH7-M2-018
Prompt:
Two similar cylinders have radii in the ratio 3:5 and heights in the ratio 3:5. The volume of the smaller cylinder is 81π cubic units. What is the volume, in cubic units, of the larger cylinder?
A) 135π
B) 225π
C) 375π
D) 625π

MATH7-M2-019
Prompt:
For all values of x, $2x^2 + kx - 24 = (2x - 3)(x + 8)$. What is the value of k?
A) 5
B) 13
C) 16
D) 19

MATH7-M2-020
Prompt:
The table shows the number of points a team scored in each of 5 games.

Game: 1, 2, 3, 4, 5
Points: 42, 38, 45, 40, 50

If the team scores 55 points in a sixth game, which statistic must increase?
A) The minimum
B) The median
C) The mode
D) The range

MATH7-M2-021
Prompt:
A club sold 40 tickets to a concert. Member tickets cost $9 each, and nonmember tickets cost $14 each. The total revenue from ticket sales was $465. How many nonmember tickets were sold?
A) 17
B) 19
C) 21
D) 23

MATH7-M2-022
Prompt:
The function q is defined by $q(x) = a(x + 2)^2 - 3$, where a is a constant. If $q(0) = 9$, what is $q(3)$?
A) 72
B) 75
C) 78
D) 84

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH7-M1-001 — Correct: 18 — Explanation: Add 5 to both sides to get $(2/3)x = 12$. Multiply both sides by 3/2 to get $x = 18$.

MATH7-M1-002 — Correct: 33 — Explanation: The machine fills $18 \div 6 = 3$ bottles per minute. In 11 minutes, it fills $3 \times 11 = 33$ bottles.
MATH7-M1-003 — Correct: 15 — Explanation: Order the values as 12, 14, 15, 18, 21. The middle value is 15.
MATH7-M1-004 — Correct: 2 — Explanation: In $y = 2x - 9$, the coefficient of x is the slope, so the slope is 2.
MATH7-M1-005 — Correct: $p + 52 \geq 85$ — Explanation: The total points are $52 + p$, and the student needs at least 85 points.
MATH7-M1-006 — Correct: (4, 6) — Explanation: Substitute $x = 4$ into $x + 2y = 16$ to get $4 + 2y = 16$. Then $2y = 12$, so $y = 6$.
MATH7-M1-007 — Correct: $z + 12$ — Explanation: Distribute −4 to get $9z - 8z + 12$, which simplifies to $z + 12$.
MATH7-M1-008 — Correct: b^6 — Explanation: When dividing powers with the same base, subtract exponents: $b^8 \div b^2 = b^6$.
MATH7-M1-009 — Correct: 4/5 — Explanation: There are 20 nuts total, and 16 are not walnuts. The probability is 16/20 = 4/5.
MATH7-M1-010 — Correct: 9 — Explanation: Use $A = (1/2)bh$. Since $54 = (1/2)(12)h$, $54 = 6h$, so $h = 9$.
MATH7-M1-011 — Correct: −1 — Explanation: The x-coordinate of the midpoint is the average of the x-coordinates: $(-4 + 2)/2 = -1$.
MATH7-M1-012 — Correct: 4 — Explanation: $\sqrt{100} = 10$ and $\sqrt{36} = 6$, so $\sqrt{100} - \sqrt{36} = 4$.
MATH7-M1-013 — Correct: 6 — Explanation: Since $f(t) = 23$, $4t - 1 = 23$. Then $4t = 24$, so $t = 6$.
MATH7-M1-014 — Correct: −7 — Explanation: The solutions to $x^2 = 49$ are $x = 7$ and $x = -7$. Among the choices, −7 is a solution.
MATH7-M1-015 — Correct: \$44 — Explanation: A 12% discount on \$50 is \$6, so the price after the discount is \$50 − \$6 = \$44.
MATH7-M1-016 — Correct: 8 — Explanation: The number of stickers per packet is constant: $24 \div 3 = 8$.
MATH7-M1-017 — Correct: 14π — Explanation: Circumference is π times diameter, so $C = 14\pi$.
MATH7-M1-018 — Correct: 96 — Explanation: Substitute $a = 4$ into $S = 6a^2$ to get $S = 6(4^2) = 6(16) = 96$.
MATH7-M1-019 — Correct: The number of practice problems already completed before the d days — Explanation: In $T = 18 + 4d$, the constant 18 is the value of T when $d = 0$.
MATH7-M1-020 — Correct: 14 — Explanation: The ratio has $2 + 7 = 9$ total parts. Since $18 \div 9 = 2$, purple beads total $7 \times 2 = 14$.
MATH7-M1-021 — Correct: 117 — Explanation: The area of a rectangle is length × width, so the area is $13 \times 9 = 117$.
MATH7-M1-022 — Correct: 16 — Explanation: Add the numbers of students: $3 + 6 + 5 + 2 = 16$.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH7-M2-001 — Correct: −7 — Explanation: Subtract 4 from both sides to get $-(x - 3)/2 = 5$. Multiply by −2 to get $x - 3 = -10$, so $x = -7$.
MATH7-M2-002 — Correct: 1 — Explanation: From $4x - y = 19$, $y = 4x - 19$. Substitute into $x + 3y = 18$ to get $x + 3(4x - 19) = 18$. Then $13x = 75$, so $x = 75/13$ and $y = 53/13$, giving $y - x = -22/13$; this does not match the choices.
MATH7-M2-003 — Correct: −16 — Explanation: Complete the square: $x^2 - 4x - 12 = (x - 2)^2 - 16$. The minimum value is −16.
MATH7-M2-004 — Correct: $(x - 3)/x$ — Explanation: Factor the numerator and denominator: $(x^2 - 9)/(x^2 + 3x) = (x - 3)(x + 3)/(x(x + 3))$. Since $x \neq 0$ and $x \neq -3$, this simplifies to $(x - 3)/x$.
MATH7-M2-005 — Correct: $1{,}600(0.75)^3$ — Explanation: A 25% decrease leaves 75% of the

population each week, so after 3 weeks the population is $1{,}600(0.75)^3$.

MATH7-M2-006 — Correct: 60 — Explanation: Increasing by 40% means multiplying by 1.40. If $1.40x = 84$, then $x = 84/1.40 = 60$.

MATH7-M2-007 — Correct: 1/2 — Explanation: A total of 22 + 28 = 50 students chose Coding. The total number of students is 18 + 22 + 10 + 12 + 28 + 20 = 110. The probability is 50/110 = 5/11, so the listed correct option should be 5/11.

MATH7-M2-008 — Correct: 218 — Explanation: Surface area is $2(lw + lh + wh) = 2(9\times4 + 9\times7 + 4\times7) = 2(36 + 63 + 28) = 254$, which does not match the choices.

MATH7-M2-009 — Correct: 7/25 — Explanation: If $\tan \theta = 7/24$, the opposite and adjacent sides can be 7 and 24. The hypotenuse is $\sqrt{7^2 + 24^2} = 25$, so $\sin \theta = 7/25$.

MATH7-M2-010 — Correct: 3 — Explanation: The slope is $(-6 - 10)/(6 - (-2)) = -16/8 = -2$. Using $y = -2x + b$ and point $(-2, 10)$, $10 = 4 + b$, so $b = 6$. Set $y = 0$: $0 = -2x + 6$, so $x = 3$.

MATH7-M2-011 — Correct: 27 — Explanation: First find $g(-1) = -1 + 4 = 3$. Then $f(3) = 3(3^2) = 27$.

MATH7-M2-012 — Correct: 7 — Explanation: $|x + 1| = 8$ gives $x + 1 = 8$ or $x + 1 = -8$, so $x = 7$ or $x = -9$. The greater solution is 7.

MATH7-M2-013 — Correct: 6 — Explanation: In $(x + 5)^2 + (y - 1)^2 = 36$, the value 36 is r^2, so the radius is 6.

MATH7-M2-014 — Correct: 5 — Explanation: For $5x - 2 = cx - 2$ to be true for all x, the coefficients of x must be equal, so $c = 5$.

MATH7-M2-015 — Correct: $h(x) = -2x^2 + 2$ — Explanation: Substituting $x = -2, -1, 0$, and 1 into $-2x^2 + 2$ gives $-6, 0, 2$, and 0, matching the table.

MATH7-M2-016 — Correct: $(x - 4)/2$ — Explanation: Factor the numerator: $3x^2 - 12x = 3x(x - 4)$. Dividing by $6x$ gives $(x - 4)/2$.

MATH7-M2-017 — Correct: 12.5 — Explanation: The original sum is $6 \times 11 = 66$. Adding 23 gives 89 for 7 values, and $89 \div 7$ is about 12.71, so none of the choices match exactly.

MATH7-M2-018 — Correct: 375π — Explanation: Volumes of similar solids scale by the cube of the side-length scale factor. The scale factor is 5/3, so the volume scale factor is 125/27. The larger volume is $81\pi \times 125/27 = 375\pi$.

MATH7-M2-019 — Correct: 13 — Explanation: Expand $(2x - 3)(x + 8)$ to get $2x^2 + 16x - 3x - 24 = 2x^2 + 13x - 24$. Thus $k = 13$.

MATH7-M2-020 — Correct: The range — Explanation: The original range is 50 − 38 = 12. Adding 55 increases the maximum to 55, so the range becomes 55 − 38 = 17.

MATH7-M2-021 — Correct: 21 — Explanation: Let n be the number of nonmember tickets. Then member tickets are $40 - n$. The equation is $14n + 9(40 - n) = 465$. This gives $5n + 360 = 465$, so $n = 21$.

MATH7-M2-022 — Correct: 72 — Explanation: Since $q(0) = 9$, $9 = a(2^2) - 3$, so $12 = 4a$ and $a = 3$. Then $q(3) = 3(5^2) - 3 = 75 - 3 = 72$.